



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY



Chandubhai S Patel Institute of Technology

CHAMOS Matrusanstha Department of Mechanical Engineering

ME – 244 MANUFACTURING PROCESSES – I LABORATORY MANUAL



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ME – 244 MANUFACTURING PROCESSES – I	
CO1	Acquire basic knowledge of manufacturing processes.
CO2	Develop ability to forecast the composition of metal/alloy obtain from furnaces.
CO3	Acquire basic knowledge of different furnaces.
CO4	Acquire basic knowledge of various casting processes and to analyze casting defects.
CO5	Understand the basics of welding processes.
CO6	Understand the different types of welding processes in depth and analyze the defects by penetration test.
CO7	Impart the knowledge of advanced welding processes.

List of Experiment (ME – 244 MANUFACTURING PROCESSES – I)		
Sr. No.	Title	Course Outcomes
1	Moulding Practice in Sand Casting	CO1
2	Tests Performed on Moulding Sand	CO1
3	Design of Gating system & Casting Simulation	CO1, CO2
4	Laboratory Demonstration of Melting and Pouring	CO2, CO3
5	Casting Defects, their Causes and Remedies	CO4
6	Moulding Practice in Investment Casting	CO4
7	Study different Welding Processes, Weld Joint Design as per I.S. code and Weld Symbols	CO5
8	Gas Welding and Gas Cutting Processes	CO6

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9	MIG & TIG Welding Process	CO6
10	Resistance Welding Process	CO6
11	Weld Defects and Dye Penetrate (DP) Test	CO6
12	Friction Welding Process	CO7

List of Assignment Experiment (ME – 244 MANUFACTURING PROCESSES – I)		
Sr. No.	Title	Course Outcomes
1	Assignment 1- Metal casting- Patterns, Sand Mould, Gating System, Defects	CO1, CO4
2	Assignment 2- Special Casting Processes	CO4
3	Assignment 3- Melting Practice	CO2, CO3
4	Assignment 4- Metal Joining Processes	CO5, CO6
5	Assignment 5- Advanced Welding Processes	CO7

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Date:

EXPERIMENT: 1
Moulding Practice in Sand Casting

AIM: To learn how to make sand mix and sand mould.

OBJECTIVE:

- To study about different types of patterns and its materials
- List different proportions of mould ingredients
- Demonstrate preparation of sand systems.
- Identify parting line.
- Demonstrate placement of pattern
- Understand ramming of sand around pattern.
- Demonstrate preparation of mould.

THEORY:

Founding or Casting is the process of producing metal/alloy components parts of desired shapes by pouring the molten metal/alloy into a prepared mould (of that shape) and then allowing the metal/alloy to cool and solidify. The solidified piece of metal/alloy is known as Casting.

Steps involved in making a Casting:

- Make a pattern out of wood, metal, plastic etc.
- In case of sand casting, select, test and prepare the necessary sand mixture for mould and core making.
- With the help of patterns prepare the mould and necessary cores.
- Melt the metal/alloy to be cast.
- Pour the molten metal/alloy into the mould and remove the casting from the mould after the metal/alloy solidifies.
- Clean and finish the casting.
- Test and inspect the casting
- Remove the defects if any.
- Relieve the casting stresses by heat treatment.
- Again inspect the casting.

Pattern:

Pattern is mould forming tool and the replica of the object to be cast. Pattern may be constructed out of the following materials according to application.

- | | |
|-----------|---------------|
| • Wood | • Metal |
| • Plastic | • Plaster |
| • Wax | • Polystyrene |

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The different types of patterns commonly used are:

- One Piece pattern
- Loose Piece Pattern
- Cope and Drag Pattern
- Segmental pattern
- Split Pattern
- Match Plate Pattern
- Sweep Pattern
- Gated Pattern

Core:

It is an obstruction which when positioned in the mould naturally does not permit the molten metal to fill up the space occupied by the core. It is made up of sand, metal, plaster and ceramics.

Types of cores classified according to

- The state or condition of core
- The nature of core material employed
- The type of core hardening process employed
- The shape and positioned of the core

Gating System:

This term refers to all passageways through which the molten metal passes to enter the mould cavity. The gating system composed of

- Pouring Cups & Basins
- Gates
- Risers
- Sprue
- Runner

Risering System:

Riser or feeder head is a passage of sand made in the cape (mould). The molten metal rises in the feeder need after the mould cavity is filled up. There are mainly two types of risers: open riser and closed riser.

REVIEW QUESTIONS:

1. What do you understand from the term 'Manufacturing'? Explain briefly.
2. What is the importance of the manufacturing processes? Give the broad classification of the manufacturing processes.
3. List the main advantages and limitation of the casting process.
4. State the different application of the casting processes.
5. How do you classify the different tools and equipment used in the foundries?
6. Draw chart which shows each section of a foundry.
7. What is pattern? How does it differ from the actual product to be made from it? Write colour coding used for pattern.
8. Describe CO₂ molding; explain any two common methods for gassing.
9. Describe the following
 - a. Bench molding
 - b. Pit molding
 - c. Machine molding

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10. Write short note on:

- | | |
|----------------------------|-------------------------|
| a. Skin dried mold | b. Plaster mold |
| c. Air dried mold | d. CO ₂ mold |
| e. Cement bonded sand mold | f. Metallic mold |

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EXPERIMENT: 2
Tests Performed on Moulding Sand

AIM: To study about tests performed for evaluation of moulding sand properties.

OBJECTIVE:

- To study about different types of tests carried out on molding sand
- To measure the permeability and hardness number for moulding sand

THEORY:

A moulding sand is supposed to possess many properties for its efficient functioning. These properties depend upon grain shape, size, distribution & the content & types of binders' additives and moisture.

Sand test indicate the moulding sand performance and help the foundry men in controlling the properties of moulding sand.

Permeability Test:

- Permeability is that property of molding sand which permits the escape of steam and other gases generated in the mould during hot metal pouring.
- Permeability is the property of rammed sand a standard size sand specimen is first rammed by a specimen rammer and is then used in the permeability tester for finding its permeability.

$$\text{Permeability Number} = V.H. / A.P.T.$$

Where, V = volume of air passed through the specimen

H = Height of the Specimen

A = Area of the Specimen

T = Time in min. taken by the air to pass through the sand specimen.

P = Air pressure recorded by the manometer in gm/cm²

Hardness Test:

- Surface hardness is measured to check the ramming density of the actual sand mould.
- Hardness can be tested with the help of an Indentation Hardness Tester.
- The instrument resembles dial indicator, carries a spring loaded spherical indenter (plunger) which penetrates into the mould surface.
- The depth of penetration achieved by the indenter with respect to the flat surface of the tester is indicated on the dial of the instrument in terms of hardness units.
- The full hardness scale is divided in 100 units.

REVIEW QUESTIONS:

1. Explain how the grain size and shape affect the performance of foundry sand.
2. Why testing of foundry sand is necessary? What are the common tests performed on foundry sand. Explain each one in detail with necessary sketches.

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3. Explain how the shape and size of grain sand affect the permeability and green strength of molding sand.
4. What is the significance of AFS fineness number and permeability number in sand properties?
5. Explain the effect of clay content on properties of sand.
6. Determine the permeability Number if 2000cc of air takes 90 seconds to pass through a standard specimen (5.08 cm height and 5.08 cm dia.) and the manometer indicates an air pressure reading of 5 gm/cm².

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EXPERIMENT: 3
Design of Gating system & Casting Simulation

Part – I Design of Gating System

AIM: To design components of gating system.

OBJECTIVE:

- To design feeder for achieving directional solidification.
- To design sprue, runners and ingates for complete filling of mould.

THEORY:

A good gating design should ensure proper distribution of molten metal without excessive temperature loss, turbulence, gas entrapping and slags.

If the molten metal is poured very slowly, since time taken to fill the mould cavity will become longer, solidification will start even before the mould is completely filled. This can be restricted by using super-heated metal, but in this case solubility will be a problem.

If the molten metal is poured very faster, it can erode the mould cavity. So gating design is important and it depends on the metal and molten metal composition.

REVIEW QUESTIONS:

1. Explain the gating system with neat sketch? What are the elements of the gating system?
2. What important requirements an ideal gating system is expected to make?
3. What functions are performed by riser? How many types of riser are their? Describe with their advantages and disadvantages.
4. Tutorial on design of gating system.

Part – II Casting Simulation

AIM: To study about casting simulation process.

OBJECTIVE:

- To study about metal filling and solidification behaviour from simulation results.

THEORY:

CASTING SIMULATION is a powerful tool to visualize progressive solidification of molten metal inside a mould cavity. It helps in identifying hot spots, which manifest as shrinkage porosity defects. The defect is usually eliminated by connecting a feeder, which is designed to solidify later than the hot spot. It supplies liquid metal to compensate the volumetric contraction at the hot spot. Feeder is cut-off and recycled, hence its size has to be optimized to ensure high yield.

By simulating the process, we hope to predict potential defects in the casting and redesign the process to eliminate the defects, before making actual castings.

General steps involved in casting simulation:

1. Solid model a cast part and save it as a .STL file.
2. Browse and upload the casting model file.
3. Wait till the simulation results are displayed.
4. Identify hot spots. Decide feeder size and location.
5. Model the part with feeder and save as a .STL file.
6. Simulate again and check the location of hot spots.

If hot spots are not shifted inside feeders, repeat 4-6.

REVIEW QUESTIONS:

1. What is casting simulation? How it is useful in casting process?
2. Explain different casting simulation software briefly.
3. Tutorial on casting simulation.

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EXPERIMENT: 4
Laboratory Demonstration of Melting and Pouring

AIM: To Study about melting and pouring practice of low melting point metal/alloy.

OBJECTIVE:

- To know about cares to be taken for successful melting and pouring of aluminium alloy.

THEORY:

Before pouring into the mould, the metal to be cast has to be in the molten or liquid state. A furnace is used to melt the metal. Different furnaces are employed for melting various ferrous or non-ferrous metals/alloys. Heat in furnace is generated by combustion of fuel, electric arc, electric resistance etc.

Metal pouring is one of the last frontiers of cost reduction and quality improvement for much of the metal casting industry. Pouring the mould is the culmination of a long series of critical steps leading to production of a quality metal casting.

Steps in Melting and Pouring Practice:

- Lead, zinc, aluminium are metals with relatively low melting point. They can be melted easily in graphite crucible in a resistance-heated furnace.
- First set the furnace at a temperature about 150° C above the theoretical melting point of the metal.
- Weigh the metal ingot pieces and press into the graphite crucible along with necessary flux. Place the crucible inside the furnace and start the furnace.
- Once the required temperature is attained, check whether the metal has melted or not. A steel rod can be carefully inserted inside the crucible and poked through the metal layer to ensure complete melting.
- When the melt is ready, take out the crucible with metal tongs, using gloves. Degassing treatment of metal can be done with use of degassing tablet or inert gas as per requirement of specific metal.
- Measure the temperature of melt with thermocouple before pouring and assure that as it is same as required for pouring or not.

REVIEW QUESTION:

1. List down the various furnaces used for melting of ferrous and non-ferrous metal/alloys. Discuss the factors to be considered for the selection of furnace.
2. Discuss about the factors to be considered while setting pouring temperature of melt?

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EXPERIMENT: 5
Casting Defects, their Causes and Remedies

AIM: To Study about various casting defects, their causes and remedies

OBJECTIVE:

- To know about different causes for casting defect and how to eliminate casting defects.

THEORY:

Casting is a process which carries risk of failure occurrence during all the process of accomplishment of the finished product. Hence necessary action should be taken while manufacturing of cast product so that defect free parts are obtained. Mostly casting defects are concerned with process parameters. Hence one has to control the process parameter to achieve zero defect parts. For controlling process parameter one must have knowledge about effect of process parameter on casting and their influence on defect.

To obtain this all knowledge about casting defect, their causes, and defect remedies one has to be analyse casting defects. Casting defect analysis is the process of finding root causes of occurrence of defects in the rejection of casting and taking necessary step to reduce the defects and to improve the casting yield.

During the process of casting, there is always a chance where defect will occur. Minor defect can be adjusted easily but high rejected rates could lead to significant change at high cost. Therefore it is essential for die caster to have knowledge on the type of defect and be able to identify the exact root cause, and their remedies.

REVIEW QUESTION:

1. Explain common casting defects with their causes and remedies in detail.
2. Discuss case study on casting defects.

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EXPERIMENT: 6
Moulding Practice in Investment Casting Process

AIM: To Study about mould making procedure in investment casting process.

OBJECTIVE:

- To understand about process chain employed for ceramic shell mould development.

THEORY:

Investment casting, also called lost wax casting, is widely used for producing ferrous and non-ferrous metal parts. Unlike other casting processes, investment casting produces net shape parts with excellent surface finish and dimensional accuracy. This manufacturing process is ideal for applications that have relatively low production quantities (10 to 10,000 pieces) or rapidly changing product designs. Since investment casting uses expendable patterns and ceramic shells, it is excellent for complex and detailed part designs. The process manufactures intricate parts that are difficult, if not impossible, to machine, forge or cast. Examples include internal passages and ports in a valve body, curved vanes of an impeller and internal cooling channels in a turbine blade.

REVIEW QUESTION:

1. Explain complete process sequence of ceramic shell mould development with neat sketch.
2. Discuss the advantages and disadvantages of investment casting process over sand and metal die casting process.

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EXPERIMENT: 7

To Study about Welding Process, Weld Joint Design as per I.S. code and Weld Symbols

AIM: To Study about Welding Process, Weld Joint Design as per I.S. code and Weld Symbols

OBJECTIVES:

- To study and classify various joining methods.
- To compare permanent joints and temporary joints.
- To study weld mechanics and weld joint design.
- To comprehend welding symbols, basic symbols and supplementary symbols

THEORY:

There are various ways of classifying the welding and allied processes. They may be classified on the basis of source of heat (flame, arc etc.) and types of interaction (fusion welding and solid state welding).

In general, various welding and allied processes are classified as follows:

- Gas Welding Processes
- Arc Welding Processes
- Resistance Welding Processes
- Solid State Welding Processes
- Thermo Chemical Welding Processes
- Radiant Energy Welding Processes

Commonly welded base metals can be classified as

- Ferrous Metals (like Wrought Iron, Cast Iron, Alloy Steel, Stainless Steel, Cast Steel etc.)
- Non-Ferrous Metals (like Al & its Alloys, Cu & its alloys, Mg & its alloys, Ni & its alloys, Zn & its alloys etc.)

Welds can be geometrically prepared in many different ways. The five basic types of weld joints are the butt joint, lap joint, corner joint, edge joint, and T-joint (a variant of this last is the cruciform joint). Other variations exist as well—for example, double-V preparation joints are characterized by the two pieces of material each tapering to a single centre point at one-half their height. Single-U and double-U preparation joints are also fairly common—instead of having straight edges like the single-V and double-V preparation joints, they are curved, forming the shape of a U. Lap joints are also commonly more than two pieces thick—depending on the process used and the thickness of the material, many pieces can be welded together in a lap joint geometry.

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The details of a joint, which includes both the geometry and the required dimensions, are called the joint design. Just what type of joint design is best suited for a particular job depends on many factors. Although welded joints are designed primarily to meet strength and safety requirements, there are other factors that must be considered. A few of these factors are as follows:

Whether the load will be in tension or compression and whether bending, fatigue, or impact stresses will be applied

- How a load will be applied; that is, whether the load will be steady, sudden, or variable
- The direction of the load as applied to the joint
- The cost of preparing the joint

Another consideration that must be made is the ratio of the strength of the joint compared to the strength of the base metal. This ratio is called joint efficiency. An efficient joint is one that is just as strong as the base metal.

Special symbols are used on a drawing to specify where welds are to be located, the type of joint to be used, as well as the size and amount of weld metal to be deposited in the joint. These symbols have been standardized by the American Welding Society (AWS). You will come into contact with these symbols anytime you do a welding job from a set of blueprints. You need to have a working knowledge of the basic weld symbols and the standard location of all the elements of a welding symbol

REVIEW QUESTIONS:

1. Give detail classification of all welding processes and explain each one.
2. List out the practical application of the welding.
3. Explain the safety recommendation in welding.
4. List the advantages and disadvantages of a permanent joint over a temporary joint.
5. Draw and describe weld location symbol.
6. What is a bead? Explain with sketch?
7. Sketch and name the following symbols in tubular form.
 - a. Welding symbols
 - b. Basic symbols
 - c. Supplementary symbols

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EXPERIMENT: 8
Gas Welding and Gas Cutting Processes

AIM: To Study about Gas Welding and Gas Cutting Processes

APPARATUS: Gas Cylinders of Oxygen and Acetylene, Welding Torch, Hose Pipes and Safety Equipment.

OBJECTIVES:

- To study principle of gas welding, gas welding setup, major equipments and identifying various gas welding flames and their applications.

THEORY:

Gas Welding is a fusion welding process of joining metals by melting them with the heat of a gas flame obtained by the combustion of a mixture of oxygen or air and a gas. Fuel gases include acetylene, hydrogen, gas, and natural gas or liquefied petroleum gas (LPG). The gases (oxygen and fuel gas) are mixed in proper proportion in a welding torch, which carries two regulators - one for controlling the quantity of oxygen and the other for controlling the quantity of fuel gas. The mixture of oxygen and acetylene is most popularly used for gas welding and produces temperature within range of 3200°C to 3300°C which makes possible to melt and weld all common metals. A filler rod, which also melts during welding, makes the joint stronger on solidification.

In Gas Cutting operation Ferrous metal is heated in to red hot condition and a jet of pure oxygen is projected onto the surface, which rapidly oxidizes oxides having lower melting point than the metal, melt and are blown away by the force of the jet, to make a cut.

Gas Welding includes all those welding processes in which gas flame is used as a heat source for melting metals. It is further divided in three main types:

(a) Air-acetylene welding, (b) Oxy-acetylene welding and (c) Oxy-hydrogen welding.

So according to it Three basic types of oxyacetylene flames used in oxy-fuel gas welding and cutting operations:

(a) Neutral flame; (b) oxidizing flame; (c) carburizing or reducing flame

Neutral flame is used for most welding operations and is highly suited for welding mild steels and cast irons, stainless steel, copper and aluminium. The oxidizing flame is used where maximum temperature is desired or in situations where oxidizing effect is not harmful, rather

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proves beneficial. The carburizing flame is mostly used for welding aluminium, metal, stainless steel, white metal, die cast metals and several other non-ferrous metals besides the high carbon steels.

There are two welding techniques generally used:

- (1) Leftward (or Forehand) welding
- (2) Rightward (or Backhand) welding (most preferable)

REVIEW QUESTIONS:

1. Explain briefly: Gas Flame and the reactions.
2. Explain various types of flames obtained in gas welding. Differentiate between gas welding and gas cutting processes.
3. Oxidizing flame makes a roaring sound. Why?
4. State the functions of flux used in Gas welding process.
5. List the various fuel gases that are used for the oxy fuel gas welding processes, describing the application of each of them.

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EXPERIMENT: 9
MIG & TIG Welding Process

Part – I TIG (GTAW) Welding Process

AIM: To Study about TIG Welding Process

APPARATUS: Welding Torch, Tungsten Electrode & Filler Metal, Cooling Water Supply, Water and Gas Solenoid Valves, Inert gas Cylinder

OBJECTIVES:

- To study the principle of arc welding process, welding cycle & different time phases involved.
- To understand various welding parameters of TIG welding process.

THEORY:

Air during welding tends to reduce the quality of weld. Inert gases are used to keep contaminants away from contacting the metal and also to remove the contaminants from the metal surfaces as the contaminants like air, dust and metal oxides also reduce the weld quality considerably. Inert arc gas welding is faster, produces cleaner weld and can weld metals considered to be difficult or impossible to weld. A GTAW process uses a non-consumable tungsten electrode which is mounted in a special electrode welder. This holder is also design to furnish a flow of inert gas around the electrode and around the arc. Welding operation is done by striking an arc between the work-piece and tungsten electrode in an atmosphere of inert gas such as helium, argon, CO₂ etc.

A GTAW process welding set utilises suitable power sources, a cylinder of organ gas, a welding torch having connections of cable for current, tubing for shielding gas supply and tubing for water for cooling the torch. The shape of torch is characteristic, having a cap at the back end to protect the rather long tungsten electrode against accidental breakage.

OPERATION SEQUENCE:

- Weld puddle is developed due to arc action on the job.
- Welding torch is moved back.
- Filler rod is moved ahead and filler metal is added to the weld puddle.
- Filler rod is withdrawn.
- Torch is moved to the leading edge of the puddle.

REVIEW QUESTIONS:

1. Explain in detail the function of Inert Gas in GTAW.

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2. Explain the following terms in brief:
DCSP, DCRP, ACHF
3. Write a short note: GTAW Tungsten Electrode
4. Why is the tungsten inert gas welding preferred for welding aluminium plates?

Part – II MIG (GMAW) Welding Process

AIM: To Study about MIG Welding Process

APPARATUS: Power Source & Cables, Welding Torch & Wire Electrode Coiled on a Spool, Wire Feed Mechanism & Controls, Shielding Gas Cylinder, Pressure Regulator, Flow Meter

OBJECTIVES:

- To study the principle of arc welding process, welding cycle & different time phases involved.
- To understand various welding parameters of MIG welding process.

THEORY:

GMAW is performed by using direct current reverse polarity as it gives both good cleaning action and fast filler metal deposition rates. A rather high welding current is desirable as this extra current break up the globules of molten metal into fine spray. This increases the rate of transfer and gives better control of arc, so that it can be directed accurately into the weld joint.

GMAW uses consumable electrode which is fed through the electrode holder into the arc and at the same speed the electrode is melted and deposited in the weld. A small adjustable speed motor is used to remove wire from a spool and feed it into the arc. CO₂ and argon/CO₂ mixtures are often use as shielding gases for welding various types of carbon sheets compared with GTAW.

REVIEW QUESTIONS:

1. Explain the modes of material transfer in GMAW.
2. What are the differences in MIG and TIG welding?
3. Explain the characteristics of arc welding machine, viz., constant current and constant voltage. Mention applications of each of them.

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EXPERIMENT: 10
Resistance Welding Process

AIM: To Study about Resistance Welding Process

APPARATUS: Alloy Electrodes, AC Power Supply

OBJECTIVES:

- To study the principle of resistance welding process, welding cycle & different time phases involved.
- To understand various welding parameters of resistance welding process.
- To know different resistance welding processes and their applications.

THEORY:-

Resistance welding refers to a group of welding processes such as spot and seam welding that produce coalescence of faying surfaces where heat to form the weld is generated by the resistance of the welding current through the work pieces. Some factors influencing heat or welding temperatures are the proportions of the work pieces, the electrode materials, electrode geometry, electrode pressing force, weld current and weld time, etc. Small pools of molten metal are formed at the point of most electrical resistance (the connecting surfaces) as a high current (100–100 000 A) is passed through the metal. In general, resistance welding methods are efficient and cause little pollution, but their applications are limited to relatively thin materials and the equipment cost can be high.

Spot welding is a resistance welding method used to join two to four overlapping metal sheets which are up to 3 mm thick each. In some applications with only two overlapping metal sheets, the sheet thickness can be up to 6 mm. Two copper electrodes are simultaneously used to clamp the metal sheets together and to pass current through the sheets. When the current is passed through the electrodes to the sheets, heat is generated due to the higher electrical resistance where the surfaces contact each other. As the heat dissipates into the work, the rising temperature causes a rising resistance, and the heat is then generated by the current through this resistance. The surface resistance lowers quickly, and the heat is soon generated only by the materials' resistance.

The water cooled copper electrodes remove the surface heat quickly, since the copper is an excellent conductor. The heat in the center has nowhere to go, as the metal of the work piece is a poor conductor of heat by comparison. The heat remains in the center, melting the metal from the center outward. As the heat dissipates throughout the work piece in less than a second the molten, or at least plastic, state grows to meet the welding tips. When the current is stopped

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the copper tips cool the spot weld, causing the metal to solidify under pressure. Some coatings, such as zinc, cause localized heating due to its high resistance, and may require pulsation welding to dissipate the unwanted surface heat into the copper tips.

If excessive heat is applied, or applied too quickly, the molten area may extend to the outside, and with its high pressure (typically 30,000 psi) will escape the containment force of the tips with a burst of molten metal called expulsion. When this occurs, the metal will be thinner and have less strength than a weld with no expulsion. The common method of checking a weld is a peel test, technically called "coach peel", as expulsion weakens the material by thinning, and makes it pass the peel test easier. A better test is the tensile test, which is much more difficult to perform, and requires calibrated equipment.

The advantages of the method include efficient energy use, limited work piece deformation, high production rates, easy automation, and no required filler materials. When high strength in shear is needed, spot welding is used in preference to more costly mechanical fastening, such as riveting. While the shear strength of each weld is high, the fact that the weld spots do not form a continuous seam means that the overall strength is often significantly lower than with other welding methods, limiting the usefulness of the process. It is used extensively in the automotive industry— cars can have several thousand spot welds. A specialized process, called shot welding, can be used to spot weld stainless steel

REVIEW QUESTIONS:

1. State the principle of resistance welding. Also, explain the welding cycle of spot welding process with sketch.
2. Explain resistance welding power source.
3. List various resistance welding processes. What are the main factors to be considered in resistance welding?
4. Explain how a tube can be manufactured from a sheet by a suitable welding process.
5. Write short notes on: -
 - a. Spot welding
 - b. Projection welding
 - c. Seam welding
 - d. Upset welding

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EXPERIMENT: 11
Weld Defects and Dye Penetrate (DP) Test

AIM: To study of weld defects and carry out Dye Penetrate (DP) test on welded joint

OBJECTIVES:

- To know the types of welding defects, and its remedies.
- To study types of testing methods available, its principle and working.
- To carry out Dye Penetrate (DP) test.

THEORY:

The ideal weld should be such that adequate fusion exists between the filler metal and edge preparation together with good penetration. There are two classes of defects: (a) those due to faulty technique on the part of operator such as lack of fusion, lack of penetration, over penetration, slag lines. These are basically the faults of the welder or choice of process. (b) Those resulting from fundamental difficulties in welding operations such as cracking and porosity.

TYPES OF WELDING-DEFECTS:

DIMENSIONAL Welding-defects can be assessed by visual inspection and by measuring with simple weld gages. They derive from improper set up or by distortion which should be controlled in a proper fixture, or by a different welding sequence. In general they should be corrected by employing proper means before welding. **MISALIGNMENT** is a setup problem.

Other appearance features which may cause rejection of these Welding-defects are excessive bead convexity and reinforcement, or the opposite condition, namely considerable concavity and undersized welds. Here the welder's technique should be improved.

UNDERCUT consists in a groove formed into the base metal, adjacent to the weld bead. It derives from improper manipulation of torch or electrode. Further training and improved skill of the welder should save future performance.

CRACKS are Welding-defects never permitted, because they are seen as stress raisers, and capable to grow until fracture. Different forms and positions of cracks can hint at their origin, and should be investigated before trying to correct for their appearance. Except for cases of lack of experience of the welder, who may be unable to end a weld bead without crater cracking, other instances derive mostly from limited weld ability of the materials, and should be dealt with

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by whoever has metallurgical experience, by means of special procedures invoking pre-heat and post heat and other tricks which the welder cannot be expected to provide.

Fine cracks that cannot be seen by visual inspections are the object of specialized inspection techniques. For which **DP test** is done which is based upon capillary action, where low surface tension fluid penetrates into clean and dry surface-breaking discontinuities. Penetrate may be applied to the test component by dipping, spraying, or brushing. After adequate penetration time has been allowed, the excess penetrate is removed, a developer is applied. The developer helps to draw penetrate out of the flaw where a visible indication becomes visible to the inspector. Inspection is performed under ultraviolet or white light, depending upon the type of dye used - fluorescent or no fluorescent (visible).

REVIEW QUESTIONS:

1. List the common welding defects. Explain briefly with neat sketch, the causes and its remedies of the defects.
2. Define destructive and non-destructive testing of weld.
3. List the non-destructive testing methods; explain each of them in detail.

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EXPERIMENT: 12
Friction Welding Process

AIM: To Study about Friction Welding Process

OBJECTIVES:

- To study the principle of friction welding process
- To understand various welding parameters of friction welding process.
- To know different friction welding processes and their applications.

THEORY:-

Friction welding is one of the solid-state welding processes that use friction heat to join members. Friction welding has superior characteristics for mechanical Properties and production performance. However, this process is limited to joining only round bar materials due to the lack of positioning-control capability. Friction welding is a process in which heat generated by friction is used to create the ideal conditions for a high-integrity welded joint between similar or dissimilar metals. In its simplest form, friction welding involves holding two components in axial alignment. Friction welding is a method for making welds in which one component is rotated relative to, and in pressure contact, with the mating component to produce heat at the faying surfaces. The weld is completed by the application of a forge force during or after the cessation of relative motion.

PRINCIPLE OF FRICTION WELDING:

Figure 1 shows the principle of basic friction welding.

1. One side of the materials to make it join is fixed, and the other material is rotated by rotational speed N (around 3000 rpm).
2. The material is touched by friction pressure P_1 (Range: 40 MPa to 450 MPa) with the other material rotated, and frictional heat is caused at the weld interface. The temperature of the material rises, transform becomes possible axially by the frictional heat, and the flash is formed. At this time, the oxide film and the pollutant on the friction surface are exhausted outside the joint area with the flash.

3. The rotation is stopped with the brake after friction time t_1 passes, upset pressure P_2 is given during upset time t_2 , and a joint complete.

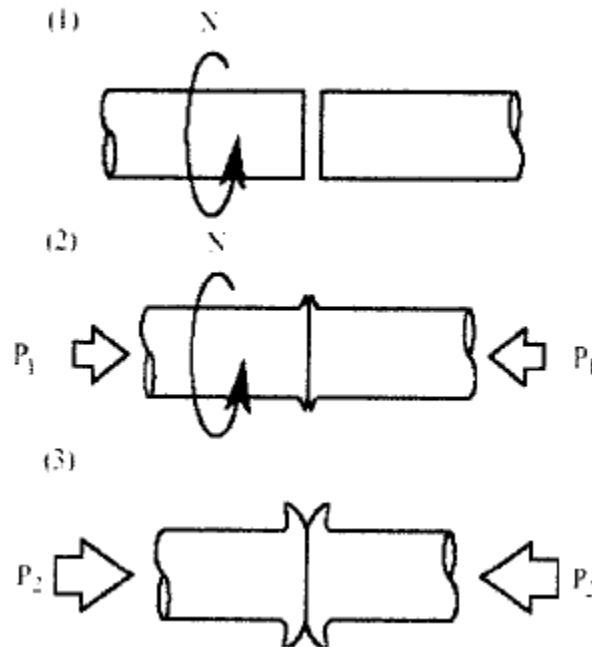


Figure 1: Basic steps in friction welding process

ADVANTAGES:

1. The edge cleaning is not required likewise in other welding process.
2. Dissimilar metals can be joined.
3. The quality of weld achieved is very high.
4. It is economical in operation.
5. Any unskilled operator can work on it since it is completely automatic in operation.

APPLICATIONS:

Because of the high quality of the weld obtained, friction welding has been widely accepted in the welding industry as well as automobile industry for welding of critical parts.

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REVIEW QUESTIONS:

1. Explain the process of friction welding?
2. State the advantages of friction welding?
3. Is it possible to use a center lathe for friction welding? Support your answer with reasons.
4. What applications would require friction welding? Why.

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